

**INNOVATIVE INFRASTRUCTURE FOR
INDUSTRIAL DEVELOPMENT: AUTOMATIC
VEHICLE PARKING ASSEMBLY
INTRODUCTION TO ENGINEERING
PROJECT REPORT**

Submitted by

Soc-A1(BATCH 1)

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**VEL TECH RANGARAJAN DR.SAGUNTHALA R&D
INSTITUTE OF SCIENCE AND TECHNOLOGY**

MAY 2024

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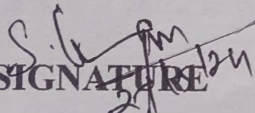
MAY 2024

Vel Tech

Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

BONAFIDE CERTIFICATE

Certified that this project Innovative Infrastructure For Industrial Development: **Automatic Vehicle Parking Assembly** is the bonafide work of **Vamsi Tarun A (VTU24588), AshwinKumar N(VTU26631), PonmuthuHarish V(VTU24343), AbidAli(VTU24497), Sai Harshith B(VTU26631), Arun Barathy(VTU24409), Arokiya Nithish J(VTU24347)** who carried out this project work under my supervision.


SIGNATURE

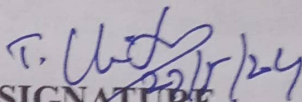
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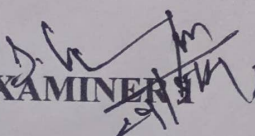
Dept. of Electronics and Communication Engineering

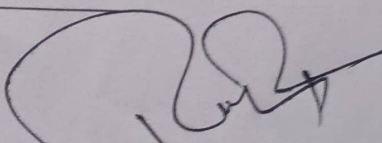
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SIGNATURE

HOD

The Examination is held on 29.5.24


EXAMINER 1


EXAMINER 2

ABSTRACT

The main aim that we have is to create a completely automated car parking system with minimal human interference. In the context of the present age, it is said that we do not have time. In our daily life, usually in the office, on the street, in shopping complexes, everywhere is the only problem, and that is the parking problem. Due to illegal parking, we have to face extreme misery every day and we all know how much time is wasted due to this disaster and that is why we have to plan this system. There has been a lot of research on this before. But we hope our ideas will be different and more perfect than others. Because, we have seen in existing systems sometimes accidents can occur in parking situations by cars going at high speed or caused by frustrated drivers unable to find a parking space for a long period of time. In our project we propose a smart and automated car parking model that will help the user in booking their parking spaces beforehand and the vehicle will be able to park automatically once in the parking zone. Hence, we aim to provide a completely safe and automated experience that is robust and can be implemented in real time and hopefully be implemented as the general norm for parking systems in the future.

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LIST OF ABBREVIATIONS

SL NO	SYMBOL	ABBREVIATIONS
1	DC	Direct Current
2	IR	Infrared

CHAPTER – I

INTRODUCTION

Autonomous vehicle, automobile that employs driver assistance technologies to remove the need for a human operator. There are six stages of automation in automobiles, ranging from fully unassisted manual driving at stage 0 to fully automated self-driving cars at stage 5. Though the terms *self-driving* or *automated* are commonly used interchangeably with *autonomous*, cars currently on the market are not capable of acting fully autonomously and cannot be operated without the intervention of a human driver. The industry standard is to use the term *automated*. The stages of automation are defined by the Society of Automotive Engineers and were adopted by the U.S. Autonomous vehicles operate by using remote-sensing technology including radar, GPS, cameras, and lidar to monitor and create a 3-D map of their environment. This environment typically includes street infrastructure, other vehicles, pedestrians, traffic lights, and road signs. Powerful computer systems process the gathered data and make decisions about vehicle operations, continually adjusting steering, cruising speed, acceleration, and braking, as sensors communicate constant changes about the vehicles' surroundings.

CHAPTER – II

EXISTING SYSTEM

For the past century or more, space for vehicle storage has been at a premium urban environments. As cars became more common around the beginning of the 20th century and the world began an urbanization trend, competition for space in cities increased.

THE FIRST AUTOMATED PARKING SYSTEMS

The Garage Rue de Ponthieu in Paris was built in 1905. A multi-level concrete structure incorporated internal elevators to move cars up and down to and from the various levels where attendants parked the cars. The elevators saved the space that many garages now occupy with ramps. In the 1920s, a system called the Paternoster became popular. It was a sort of Ferris wheel type of arrangement occupying the footprint of two cars, but usually with the capacity of eight cars. A chain was routed over sprockets at the top of the structure and under the floor, and compartments attached to the chain each held one vehicle. The compartments became accessible at the floor level in succession as the chain rotated over the sprockets. About the same, a system called the Kent Automatic Garages gained popularity. In this arrangement, an electric “parker” lifted the rear wheels of a car and moved it to or from an available space using an elevator to move among levels in the garage. One of the first of these was installed in New York in 1928.

CHAPTER – III

PROBLEM IDENTIFICATION

Current smart parking systems aim to make finding a spot easier, but they aren't without their issues. Here's a closer look at some of the problems:

Limited impact on overall parking availability: While they help locate empty spaces, they don't create more. This can be a major shortcoming in areas with chronic parking shortages. Even with increased efficiency, there might simply not be enough spaces for everyone.

Tech dependence and cost: These systems often rely on sensor networks to track occupancy. Installing and maintaining these sensors can be expensive, and their effectiveness depends on a reliable data connection. If the sensors malfunction or the network goes down, the real-time data that's key to the system can be disrupted.

Data security concerns: Smart parking systems collect data on vehicle location and parking duration. Ensuring this data is secure from hackers or breaches is crucial, especially with the growing importance of data privacy.

Standardization roadblocks: Different cities or regions may have varying parking regulations and enforcement methods. This lack of uniformity can make it difficult to implement a single, widespread smart parking solution that works everywhere.

CHAPTER – IV

PROPOSED SYSTEM

The model describes about a unique automatic car parking system in which the car is parked automatically by the automation mechanism consisting of a centralized lift which is attached with a conveyer belt and a sheet above it in order to place a vehicle in a specified location. The series is sequential as a floor should be fully filled with vehicles and further it is moved to another floor. For checking whether the car is parked, it is found using IR sensor. The centralized lift is lifted using dc motor and will be stopped at a point in order to place the vehicle using a pulley system.; the code will be given for each and every condition till the vehicle is parked at the location. The vehicles are not parked at random but parked sequentially in order to balance the area and load of each and every floor. This parking system consists of many slots where car is parked sequentially floor by floor. Automatic car parking system makes car parking system with visual use of IoT with Arduino, and IR sensor for its implementation. feature provided: Provides feature for checking the availability of the parking slot at the parking place using IoT.

CHAPTER – V

BLOCK DIAGRAM

Fig.5

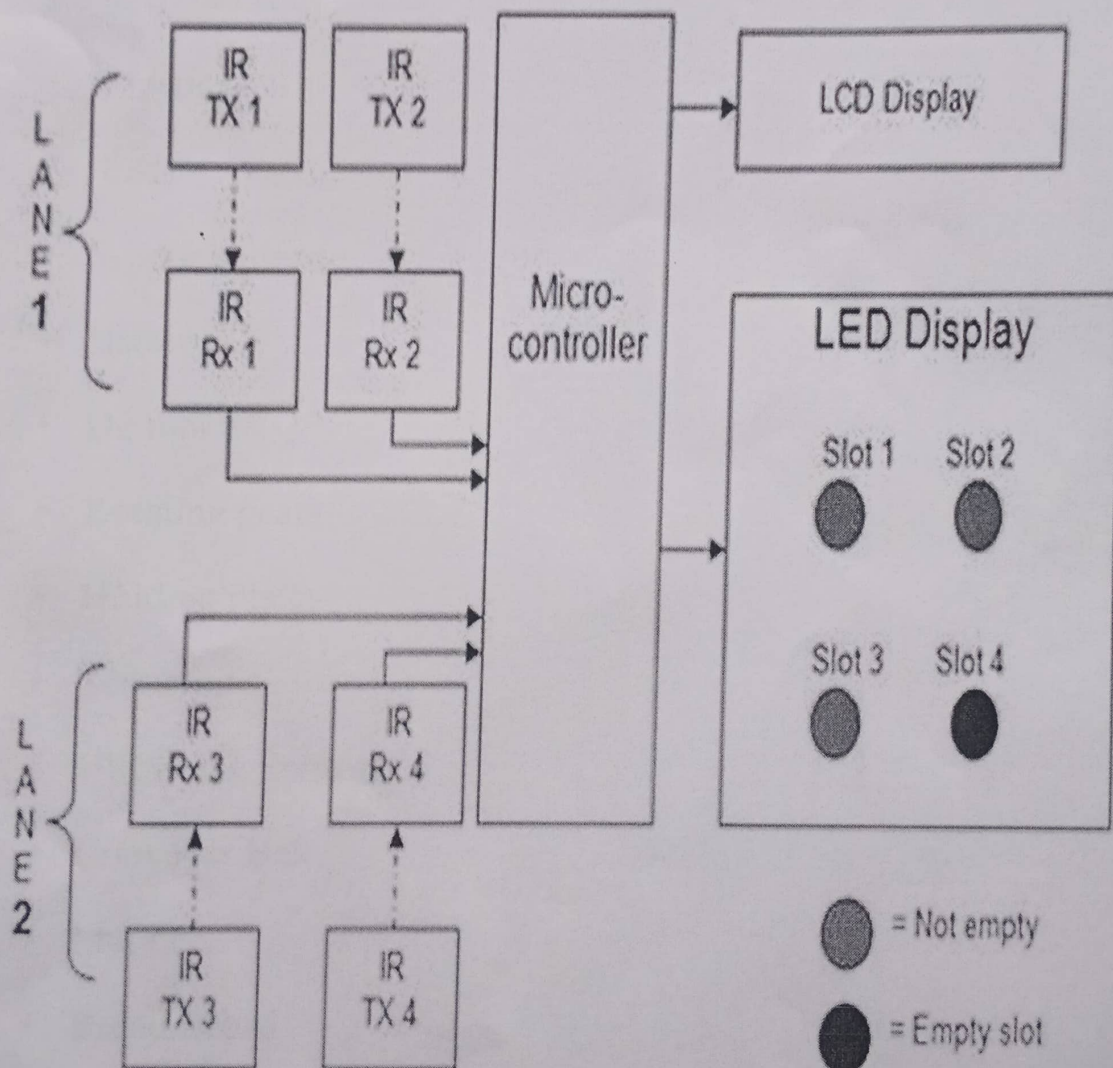


FIG.5.1 block diagram

CHAPTER – VI

MATERIALS

- Connecting wires
- pipe
- Arduino Uno
- Plastics
- Car
- Wooden piece
- Battery
- Dc motors
- Rotating gears
- Holding plates
- Screw pins
- Ultrasonic Sensors
- Conveyer Belt
- Magnets
- Bread Board
- Servo motors

CHAPTER - VII

CIRCUIT/PROTOTYPE DESIGN

AUTOMATIC VEHICLE PARKING ASSEMBLY



FIG.7.1 Prototype Model of our project

CHAPTER – VIII

WORKING

- Step 1: rotate the base by 360° and stick on to the position (by magnets) by colour identification of the specific place
- Step 2: Create lift by using pipe and motor which is like climbing method by rotating wheels fitted in motors on the pipe which will helps to climb
- Step 3: The IR sensor fitted in lift will helps to find the respective floor
- Step 4: Next step is to create the body of structure or outer structure
- Step 5: To place the car in correct position we need to use treadmill like structure to pass car from lift to parking lot
- Step 6: we need to fit IR sensor to find out vacant parking space
- Step 7: After finding we need to load directions to reach parking space by Arduino

CHAPTER – IX

SOFTWARE DETAILS

- Github
- Arduino IDE

GITHUB

GitHub is a developer platform that allows developers to create, store, manage and share their code. It uses Git software, providing the distributed version control of Git plus access control, bug tracking, software feature requests, task management, continuous integration, and wikis for every project. Headquartered in California, it has been a subsidiary of Microsoft since 2018. It is commonly used to host open source software development projects. As of January 2023, GitHub reported having over 100 million developers and more than 420 million repositories, including at least 28 million public repositories. It is the world's largest source code host as of June 2023.

ARDUINO IDE

Arduino IDE is an open-source software, designed by Arduino.cc and mainly used for writing, compiling & uploading code to almost all Arduino Modules. It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process. It is available for all operating systems i.e. MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role in debugging, editing and compiling the code.

CHAPTER – X

SOURCE CODE

```
#include <Servo.h>

#include <Wire.h>

// Define pin numbers for components

#define liftMotorPin1 2

#define liftMotorPin2 3

#define conveyorMotorPin1 4

#define conveyorMotorPin2 5

#define liftUltrasonicTrigPin 6

#define liftUltrasonicEchoPin 7

#define liftRotationServoPin 8

#define conveyorUltrasonicTrigPin 9

#define conveyorUltrasonicEchoPin 10

#define parkingIRSensorPin 11 // Single IR sensor for car entry and exit


// Define pins for the color sensor

#define S0 12

#define S1 13

#define S2 A0
```

CHAPTER - XI

ADVANTAGES AND DISADVANTAGES

ADVANTAGES OF AUTOMATED PARKING SYSTEM

1. **Saves time finding parking** – Automated Parking Systems make it quicker to park your car, as they eliminate the need to hunt for a spot.
2. **Reduces car theft risk** – There's a lower chance of car theft, since the system is tightly controlled and monitored.
3. **Increases parking space efficiency** – These systems use space more efficiently than traditional parking, fitting more cars into the same area.
4. **Minimizes parking-related accidents** – They also help to prevent accidents, as cars are parked mechanically without human error.

DISADVANTAGES OF AUTOMATED PARKING SYSTEM

5. **Reduces car theft risk** – There's a lower chance of car theft, since the system is tightly controlled and monitored.
6. **Increases parking space efficiency** – These systems use space more efficiently than traditional parking, fitting more cars into the same area.
7. **Minimizes parking-related accidents** – They also help to prevent accidents, as cars are parked mechanically without human error.
8. **Lessens vehicle emissions** – Finally, they cut down on car emissions, as less time spent driving around means less pollution.

CHAPTER – XII

HARDWARE DISCUSSION

SERVO MOTOR:



FIG 8.1 servo motor

A servo motor is defined as an electric motor that allows for precise control of angular or linear position, speed. It consists of a suitable motor coupled to a sensor for position feedback and controller that regulates the motor's movement according to a desired setpoint.

DC MOTOR:

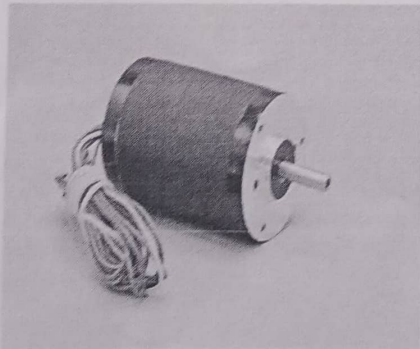


FIG 8.2 Dc Motor

A magnetic field arises in the air gap when the field coil of the DC motor is energised. The created magnetic field is in the direction of the radii of the armature.

MOTOR DRIVER:

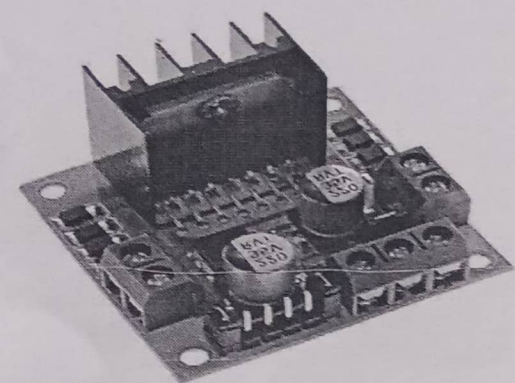
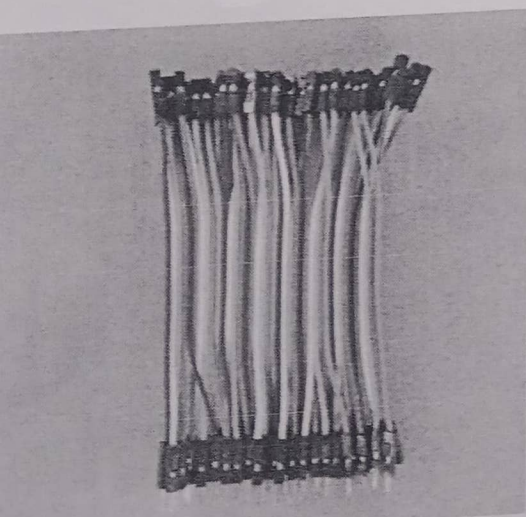


FIG 8.3 Motor Driver

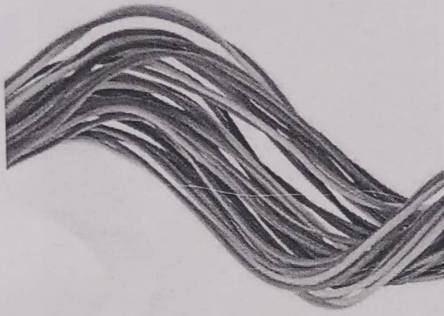
Motor drives (also known as drive systems or simply drives) are devices that control the speed, torque, and direction of an electric motor. These are used in a myriad of applications – from industrial machinery to domestic appliances.

JUMPER WIRES:



Jumpers are tiny metal connectors used to close or open a circuit part. They have two or more connection points, which regulate an electrical circuit board. Their function is to configure the settings for computer peripherals, like the motherboard. Suppose your motherboard supported intrusion detection. A jumper can be set to enable or disable it.

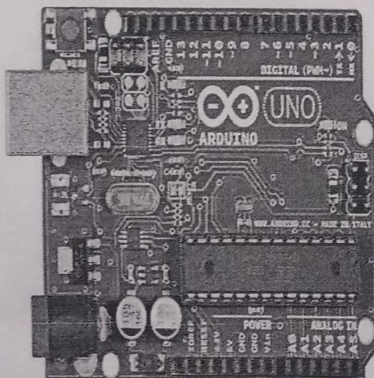
CONNECTING WIRES:



Wires are pieces of metal that transport electricity. They are usually flexible which makes them easier to use. These electrical conductors are key to all electrical devices, from the electric circuit board in a computer to the transformer to hundreds of kilometers..

FIG 8.5 Connecting Wires

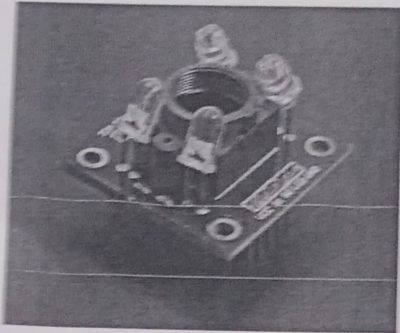
ARDUINO UNO:



Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs – light on a sensor, a finger on a button, or a Twitter message – and turn it into an output – activating a motor, turning on an LED, publishing something online.

FIG 8.6 Arduino UNO

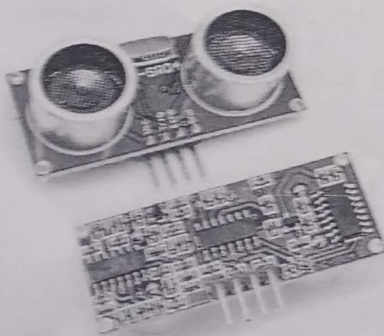
COLOUR SENSOR:



A color sensor is a type of “photoelectric sensor” which emits light from a transmitter, and then detects the light reflected back from the detection object with a receiver. A color sensor can detect the received light intensity for red, blue and green respectively.

FIG 8.6 Colour Sensor

IR AND ULTRASONIC SENSOR:



IR sensors (Infrared sensors) are electronic devices that: Measure and detect infrared radiation in their surrounding environment.

Ultrasonic sensors are electronic devices that: Measure the distance to a target by emitting ultrasonic sound waves and measuring the time between emission and reception.

FIG 8.7 IR And Ultrasonic Sensor

CHAPTER – XIII

CONCLUSION

Automated car parking systems (APS) offer a variety of advantages over traditional parking garages. They can significantly increase parking capacity by stacking cars vertically and eliminating the space needed for driver maneuvering. Additionally, they can improve convenience and safety by removing the need for drivers to park themselves, especially in tight spaces.

However, APS also has some drawbacks. They can be expensive to install and maintain, and they may not be suitable for all types of vehicles. Additionally, some people may be concerned about the safety of entrusting their car to a mechanical system.

Overall, automated car parking systems are a promising technology that can help to address the growing demand for parking in urban areas. As the technology continues to develop, APS systems are likely to become more affordable and reliable, making them a more attractive option for both drivers and property owners

CHAPTER – XIV

REFERENCES

SL NO	TITLE	NAME OF JOURNAL/PUBLICATION	AUTHOR	YEAR
1	Auto evolution	How automated parking systems work	Patrasu Daniel	2012
2	The Transportation Research Forum	CARS, Parking and sustainability	Sanders Medonald	2013
3	Automated parking system: status in united states	The original	Richard. S	2012
4	The automation dilemma	The original	Labds	2011
5	Automated parking world	The New York times	Gary	2003
6	Road to robotic parking	Urban land	Frances	2015
7	Waiting for the robotic garage	Urban land	Jack	2012
8	Green and save	Green garages	Staff	2012
9	Is it right for u for automated parking?	Parking today	Cudnes	2003
10	Man vs Machine	International parking institute	Don	2012